

91259 - ARCHITECTURES AND PLATFORMS FOR ARTIFICIAL INTELLIGENCE (Modulo 2) - 6 cfu

93398 - ARCHITECTURES FOR ARTIFICIAL INTELLIGENCE* M - 6 cfu

*Componente del corso integrato HARDWARE SOFTWARE DESIGN OF AI AND IOT SYSTEMS M I.C

AA 2024-25

Angelo Garofalo -DEI, Moreno Marzolla -DISI (Modulo 1), LM Artificial Intelligence, Francesco Conti -DEI (Modulo 2), LM Ingegneria Elettronica

<http://eyeriss.mit.edu> (slides) and:

Stanford CS231, ETHZ ML on MCUs, Hot Chips tutorials, ISCA 2017, 2020, ACACES 2020, more...

Texts: <https://www.deeplearningbook.org/>, <https://mml-book.github.io/book/mml-book.pdf>,
<http://d2l.ai/index.html> (online interactive book with exercises)
<https://link.springer.com/book/10.1007/978-3-031-01766-7> (HW focus)



Class Content & Organization

Module 1: Tutor – Davide Nadalini (+Lorenzo Lamberti, Luka Macan, Alberto Dequino)

DATE	TIME	WHERE	WHO	WHAT
17/09	14-17	0.1	Garofalo	Intro to ML (1h) + Thesis presentations (2h)
20/09	14.30-17.30	4.1	Garofalo	Linear Classifier. Backpropagation & Training
24/09	14-17	0.1	Garofalo	Convolutional Neural Networks 1
27/09	14.30-17.30	LAB1	Nadalini	L1: DNN Definition & Training
01/10	14-17	0.1	Garofalo	Convolutional Neural Networks 2
11/10	14.30-17.30	LAB1	Nadalini	L2: Embedded Programming
15/10	14-17	0.1	Garofalo	Quantization & Pruning
18/10	14.30-17.30	LAB1	Nadalini	L3: Shrinking and Quantization
22/10	14-17	0.1	Garofalo	Execution of CNNs on Edge Devices
25/10	14.30-17.30	LAB1	Nadalini	L4: Parallel programming
29/10	14-17	0.1	Garofalo	Introduction to Transformers

Exam for Module 1: Written



Miniprojects available for people willing to go deeper on some specific topics

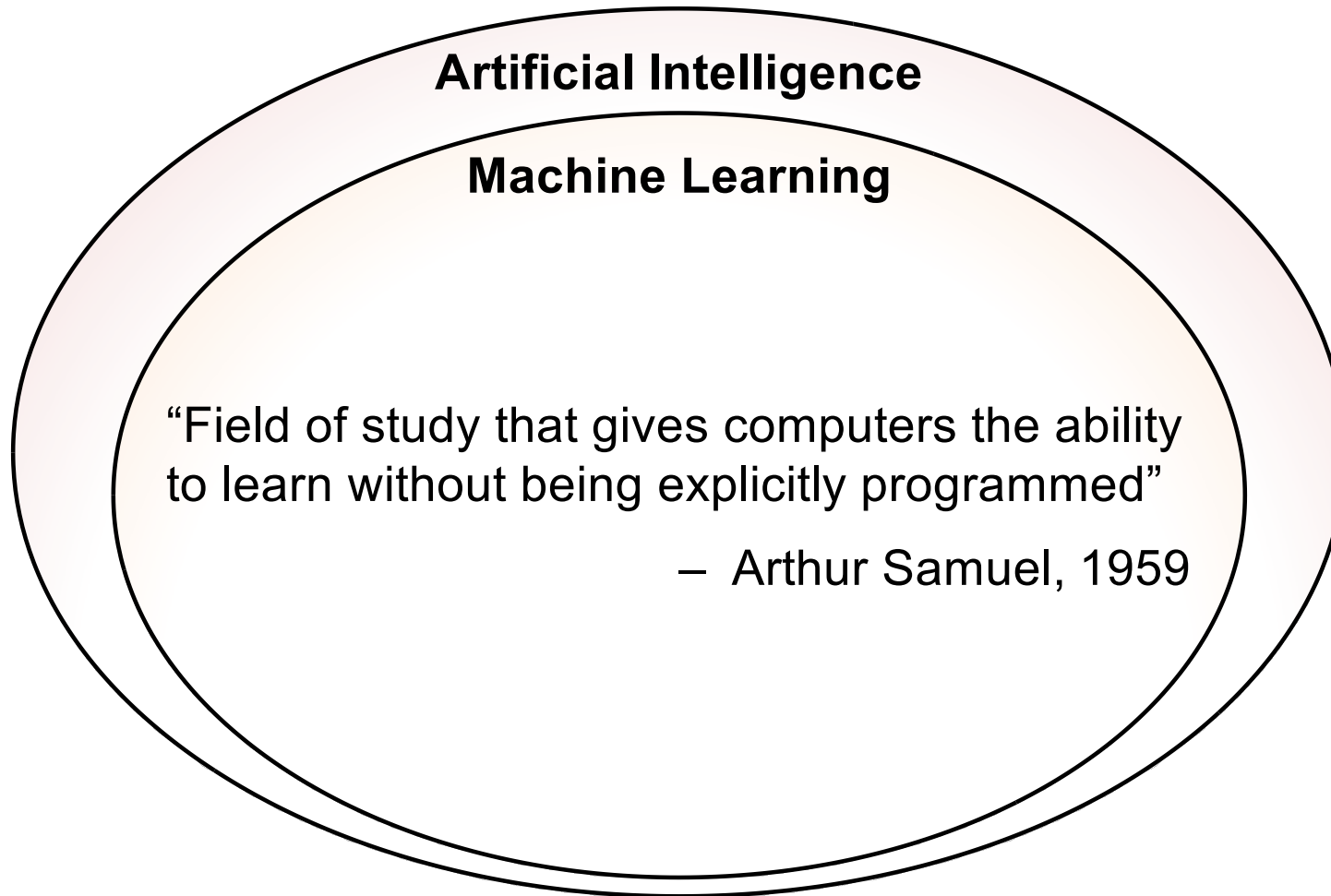
Artificial Intelligence

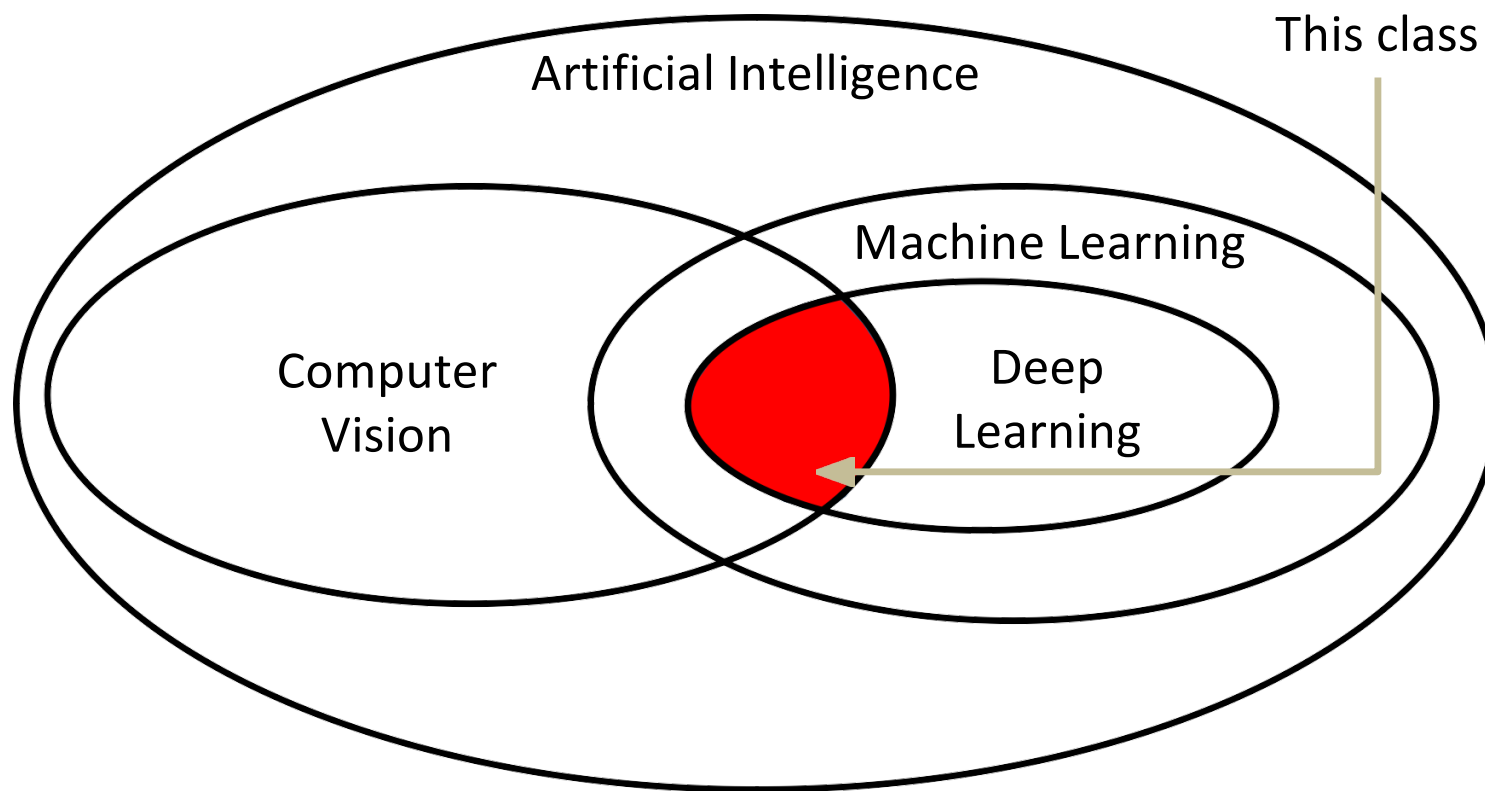
Artificial Intelligence

“The science and engineering of creating intelligent machines”

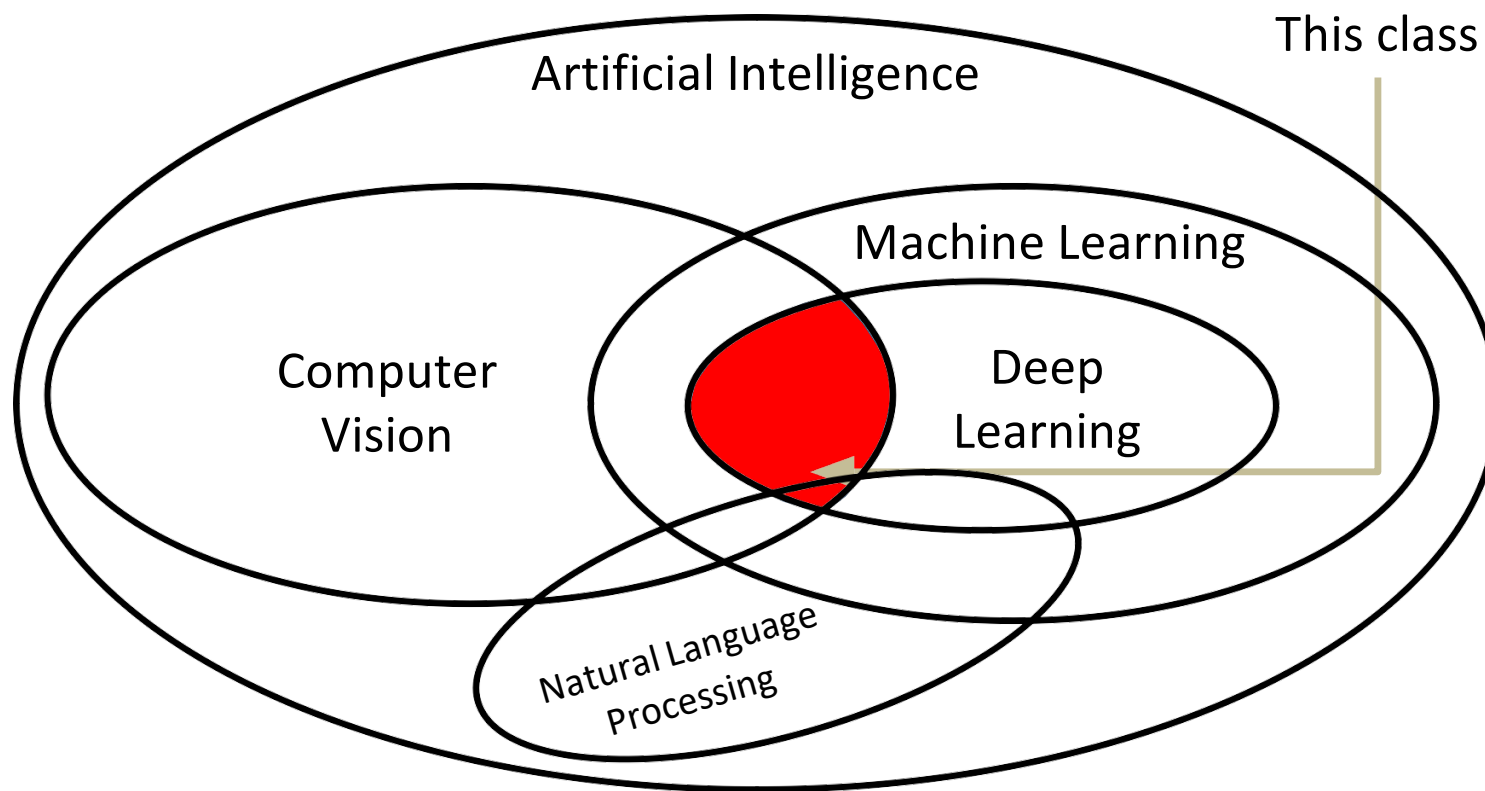
- John McCarthy, 1956

AI and Machine Learning





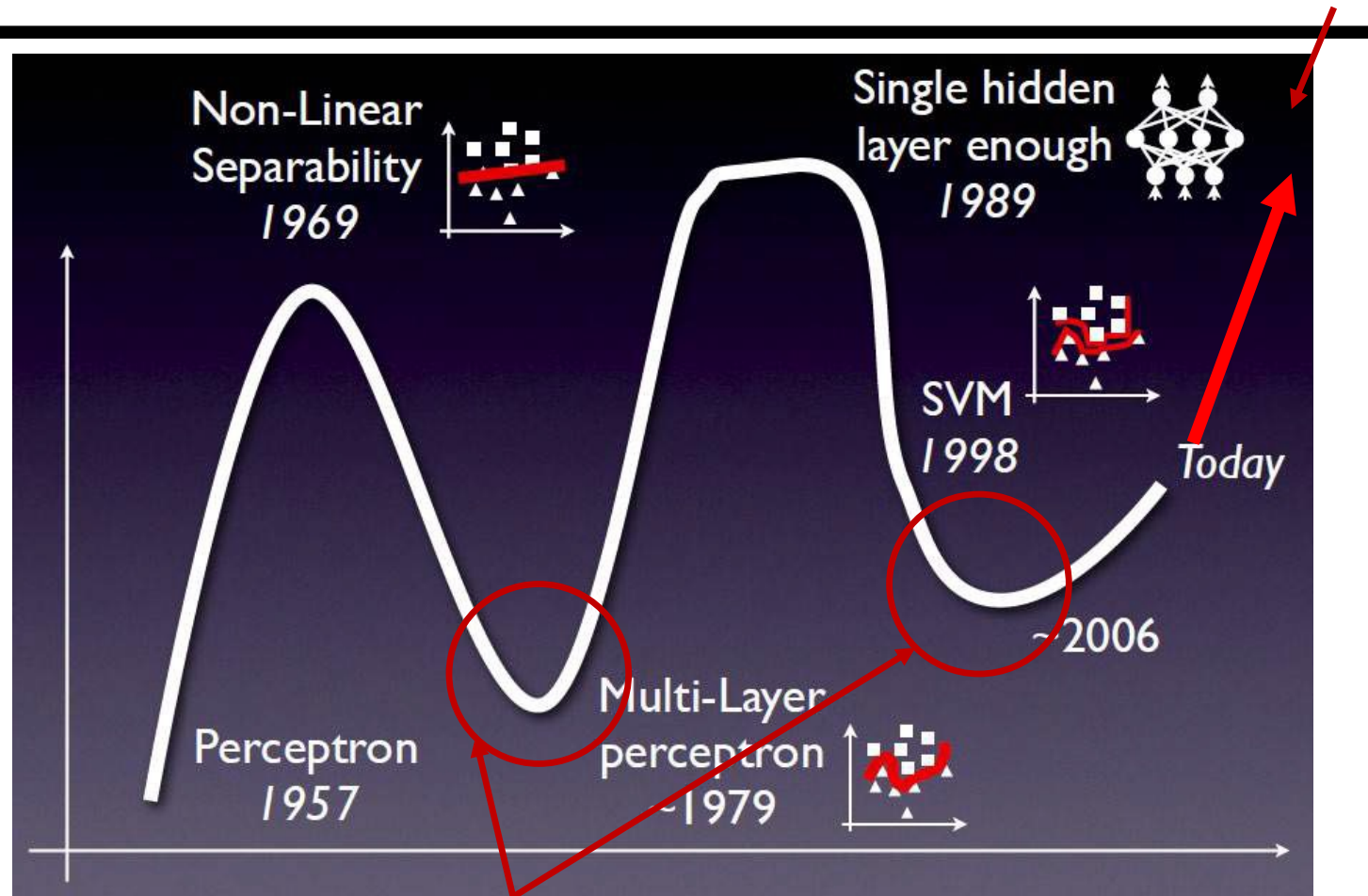
Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson

ML > NN + a bit of NN history

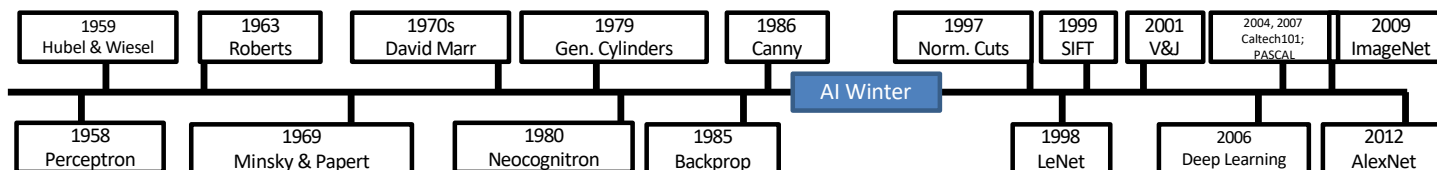
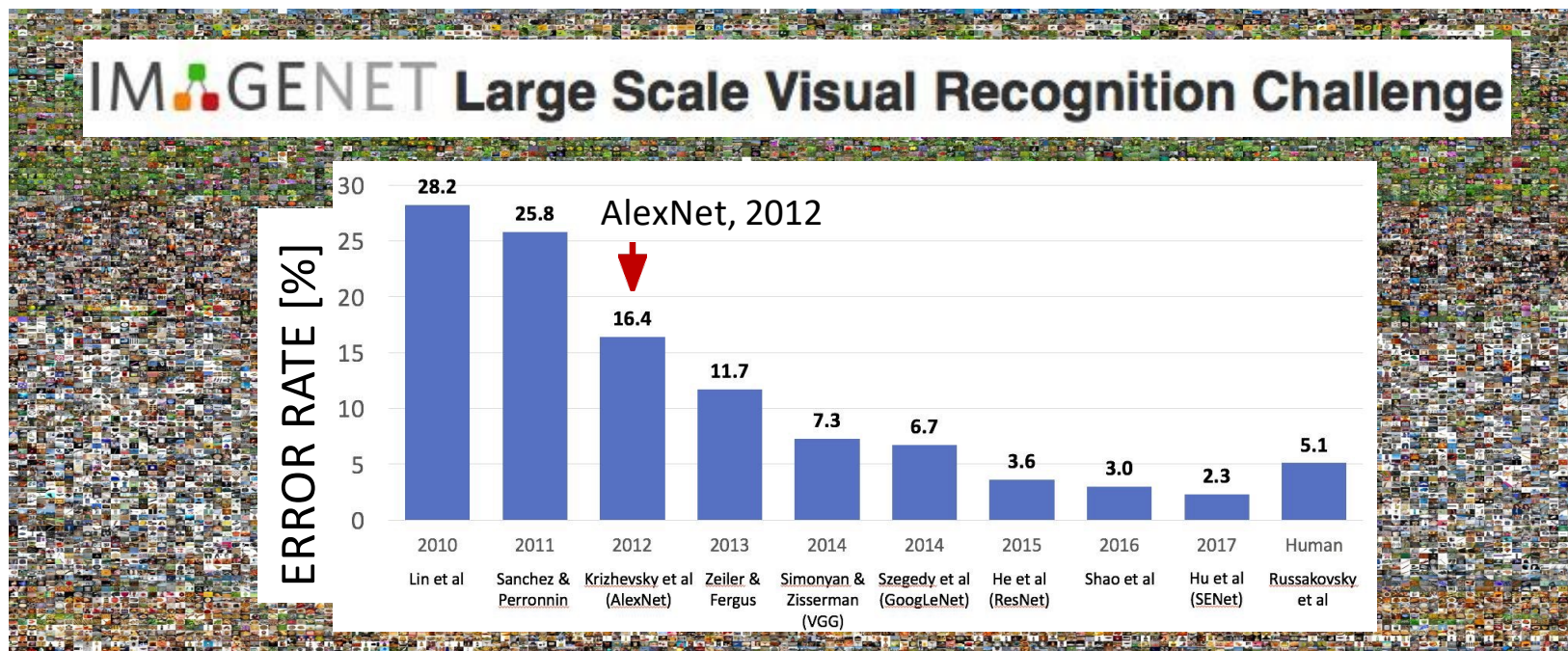
ML
“summer”



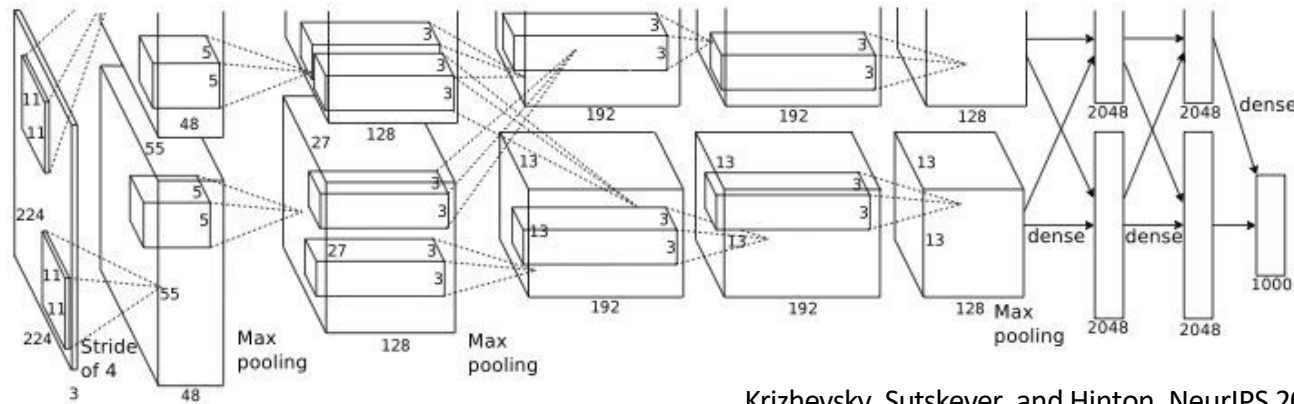
ML winters

From O. Temam keynote, ISCA'10 4

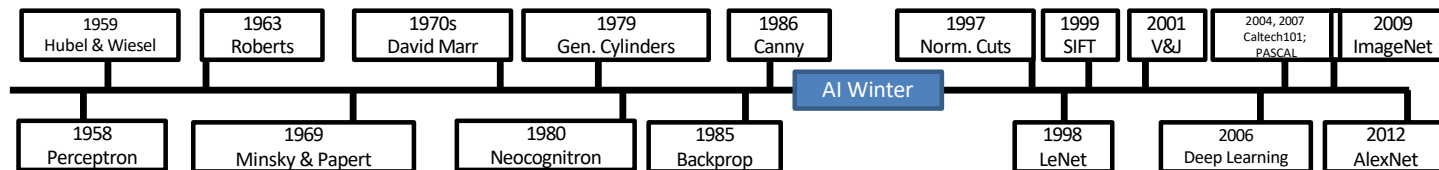
Why the explosion?



AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS 2012



Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Image Classification

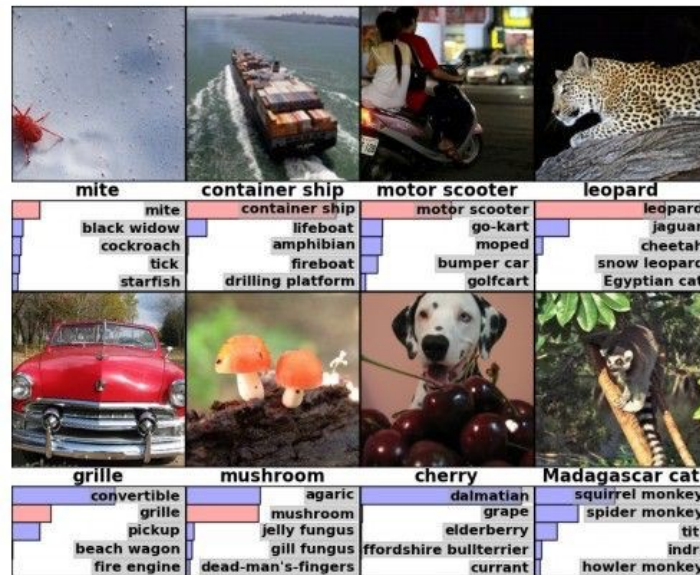


Image Retrieval

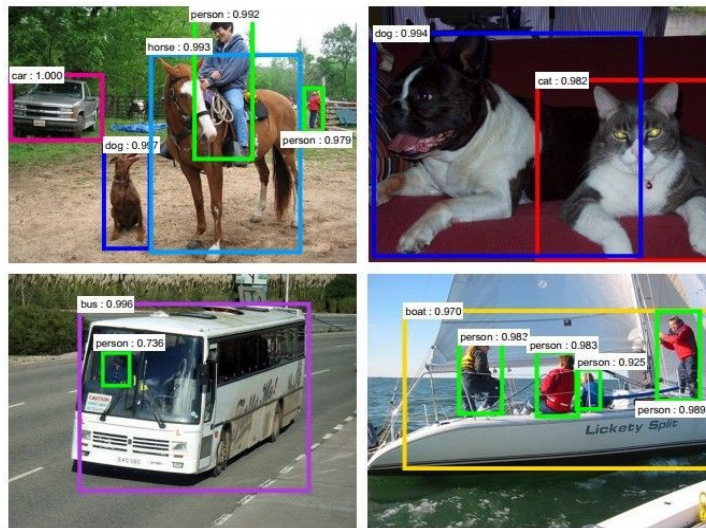


Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

Slide inspiration: Justin Johnson

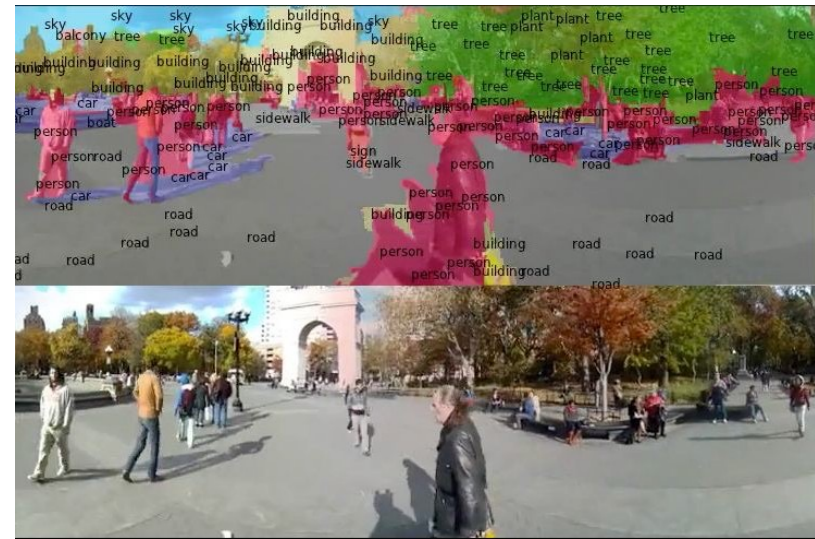
2012 to Present: Deep Learning is Everywhere

Object Detection



Ren, He, Girshick, and Sun, 2015

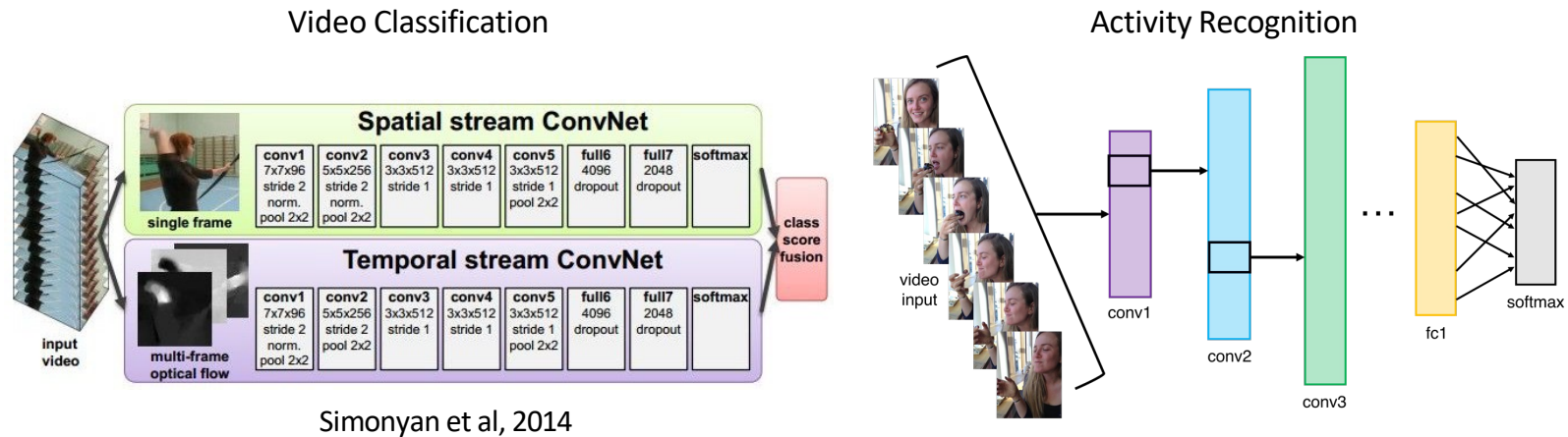
Image Segmentation



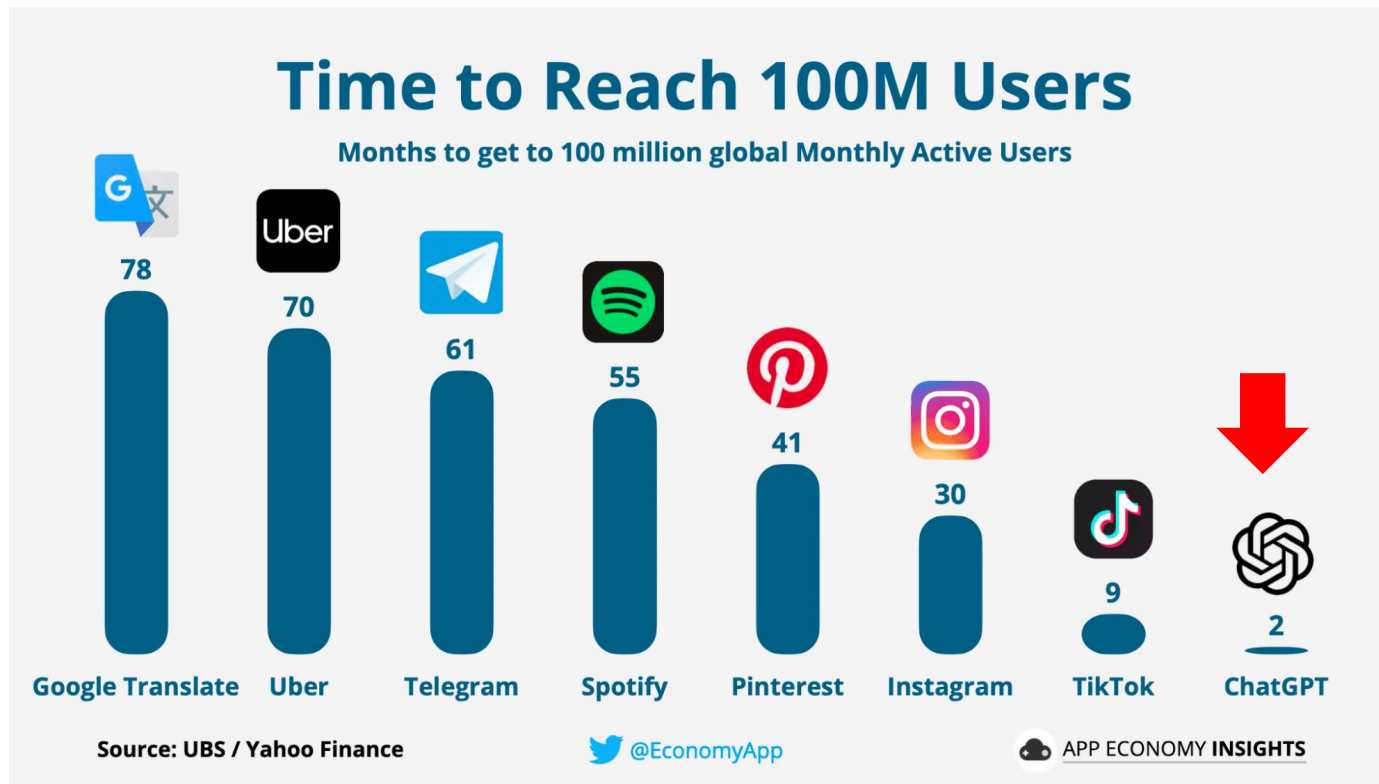
Fabaret et al, 2012

Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere



2012 to Present: Deep Learning is Everywhere

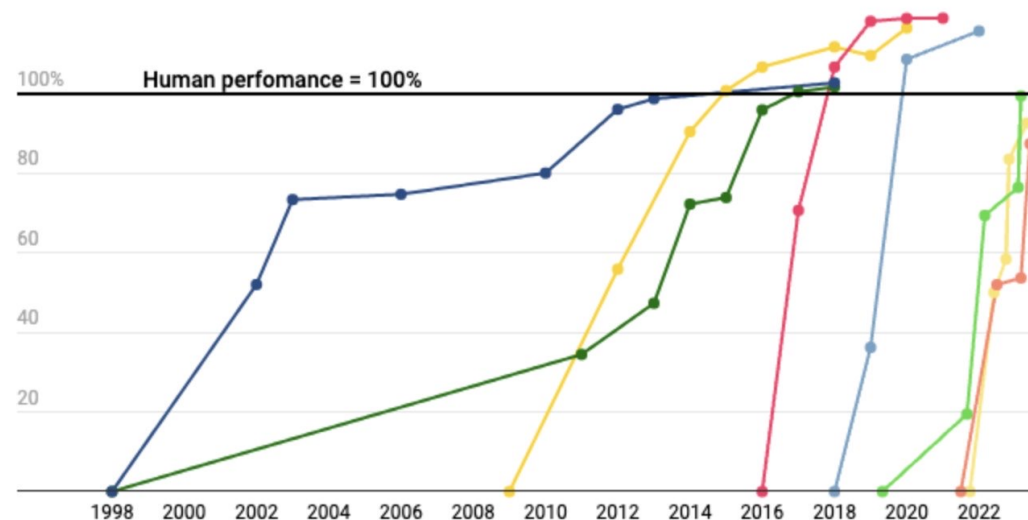


Truly everywhere!

AI has surpassed humans at a number of tasks and the rate at which humans are being surpassed at new tasks is increasing

State-of-the-art AI performance on benchmarks, relative to human performance

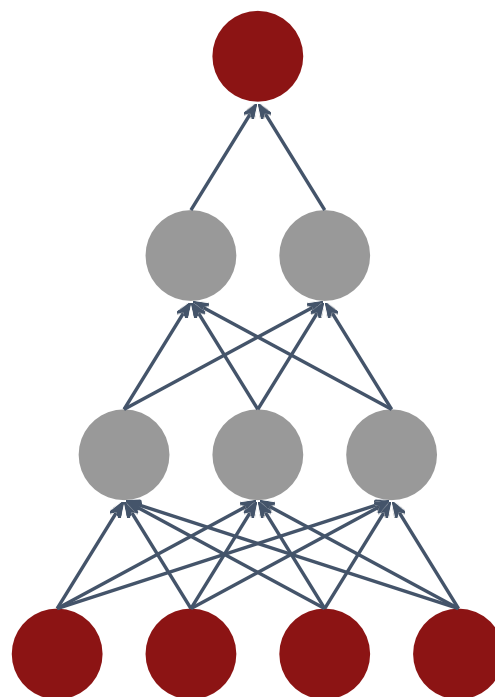
● Handwriting recognition ● Speech recognition ● Image recognition ● Reading comprehension
● Language understanding ● Common sense completion ● Grade school math ● Code generation



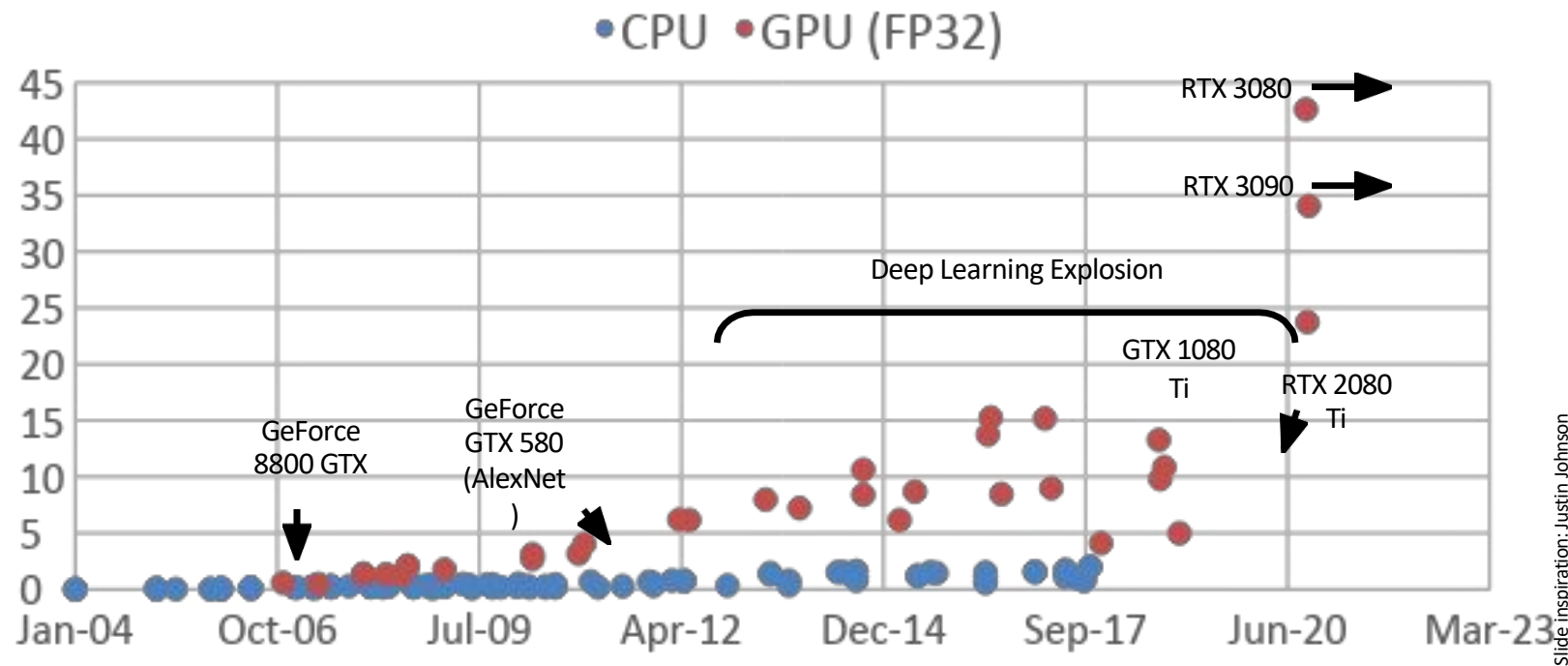
For each benchmark, the maximally performing baseline reported in the benchmark paper is taken as the "starting point", which is set at 0%. Human performance number is set at 100%. Handwriting recognition = MNIST, Language understanding = GLUE, Image recognition = ImageNet, Reading comprehension = SQuAD 1.1, Reading comprehension = SQuAD 2.0, Speech recognition = Switchboard, Grade school math = GSK&k, Common sense completion = HellaSwag, Code generation = HumanEval.

Chart: Will Henshall for TIME • Source: ContextualAI

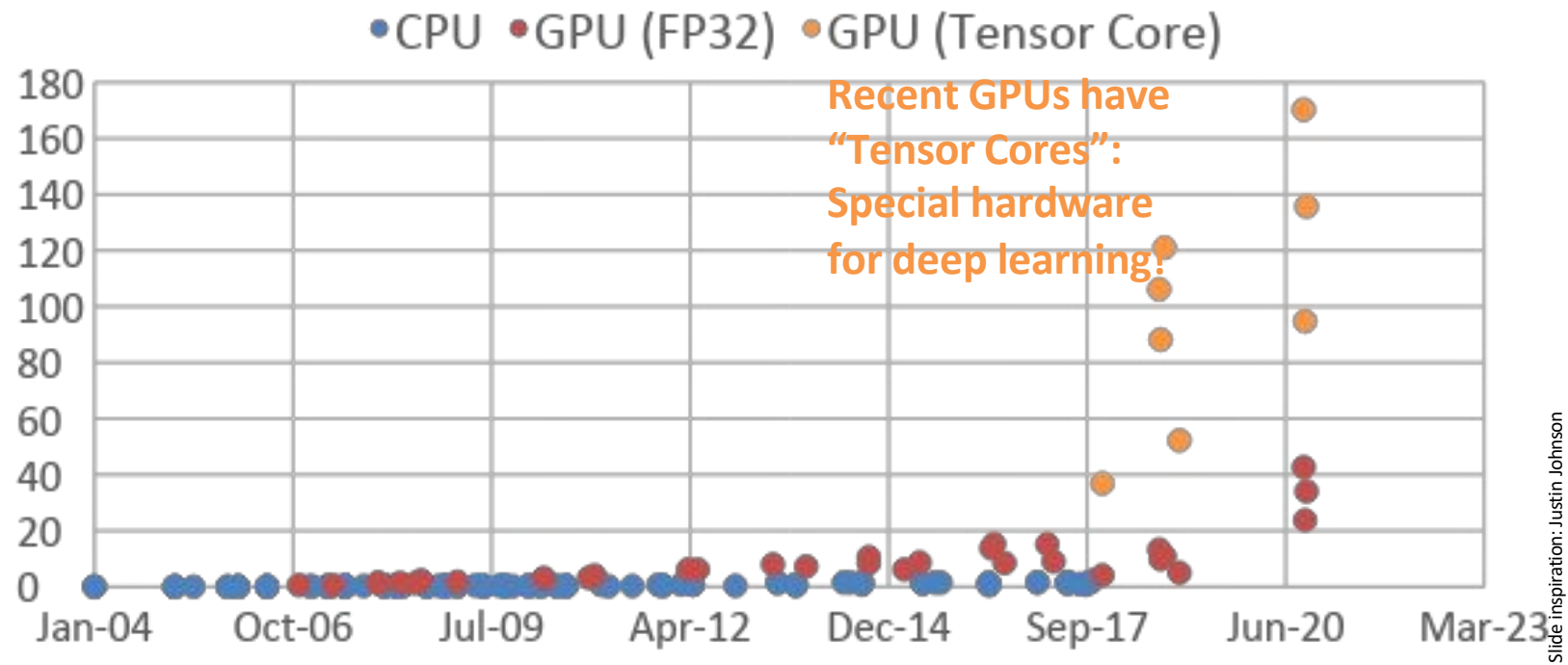
TIME



GFLOP per Dollar

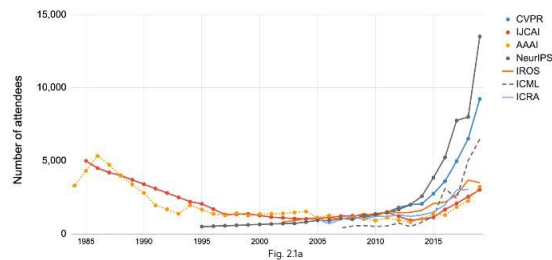


GFLOP per Dollar



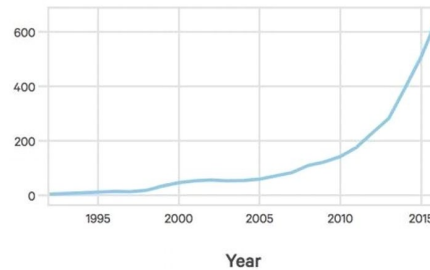
AI's Explosive Growth & Impact

Attendance at large conferences (1984-2019)
Source: Conference provided data.



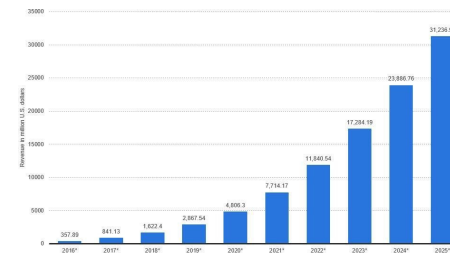
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

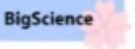
Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

The cost of Training large NNs (LLMs)

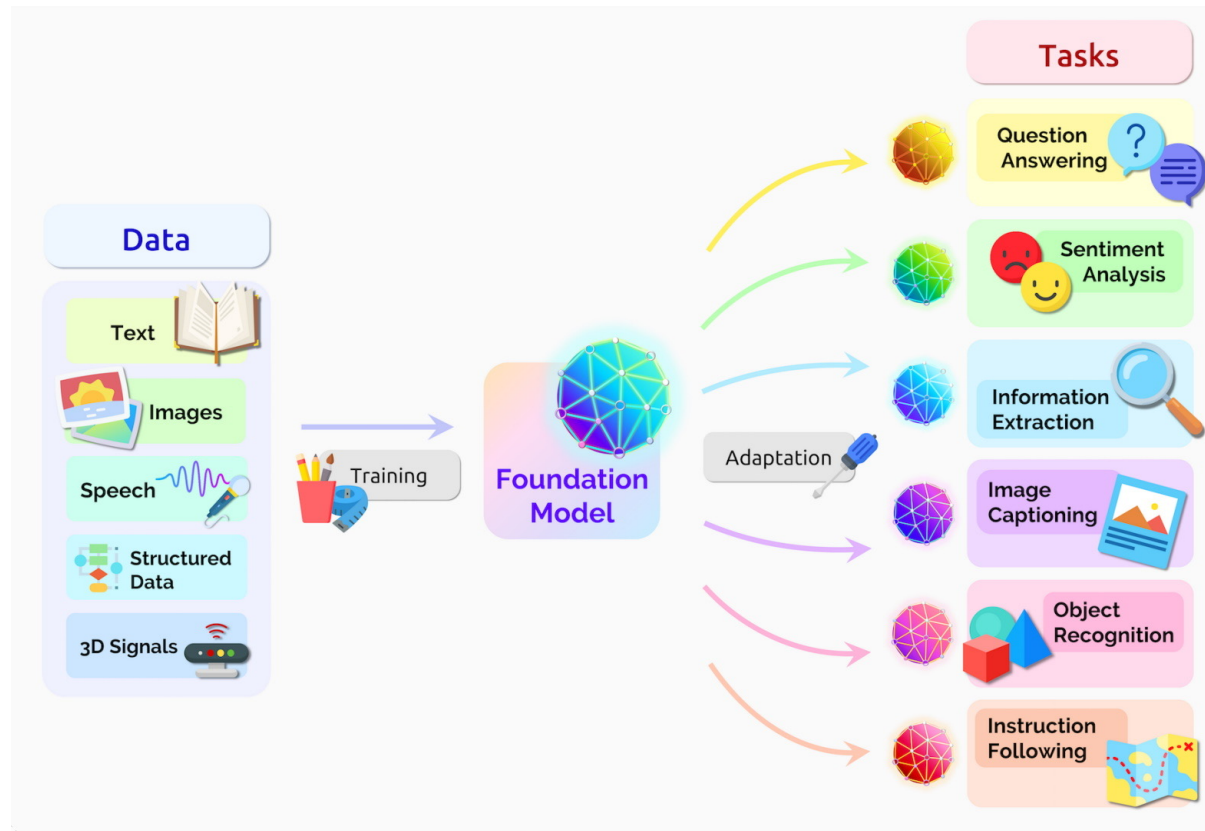
Model		GPT-3	BLOOM	LLaMA	LLaMA-2	T5	PaLM
Developer		 OpenAI		 Meta			
Model Size (# parameters)		175B	175B	7B, 13B, 33B, 65B	7B, 13B, 34B, 70B	11B	540B
Training Data (# tokens)		300B	350B	1.4T	2T	34B	795B
Training Compute (FLOPs)		3.2E+23	3.7E+23	9.9E+23	1.5E+24	2.2E+21	2.6E+24
Processor	Manufacturer	Nvidia	Nvidia	Nvidia	Nvidia	Google	Google
	Type	GPU	GPU	GPU	GPU	TPU	TPU
	Model	V100	A100	A100	A100	TPU v3	TPU v4
Processor Hours		3,552,000	1,082,990	1,770,394	3,311,616	245,760	8,404,992
Grid Carbon Intensity (kgCO2e/KWh)		0.429	0.057	0.385	0.423	0.545	0.079
Data Center Efficiency (PUE)		1.1	1.2	1.1	1.1	1.12	1.08
Energy Consumption (MWh)		1,287	520	779	1,400	86	3,436
Carbon Emissions (tCO2e)		552	30	300	593	47	271

$$KWh = \text{Hours to train} \times \text{Number of Processors} \times \text{Average Power per Processor} \times PUE \div 1000$$

$$tCO2e = KWh \times \text{kg CO2e per KWh} \div 1000$$

From: 'Reducing the Carbon Footprint of Generative AI' by Boris Gamazaychikov

The era of Foundation Models



New way of looking at NN models

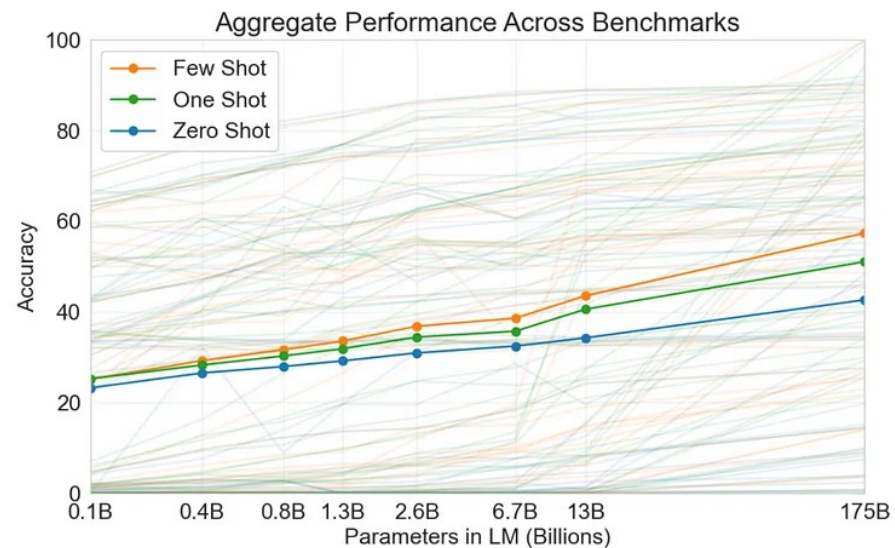
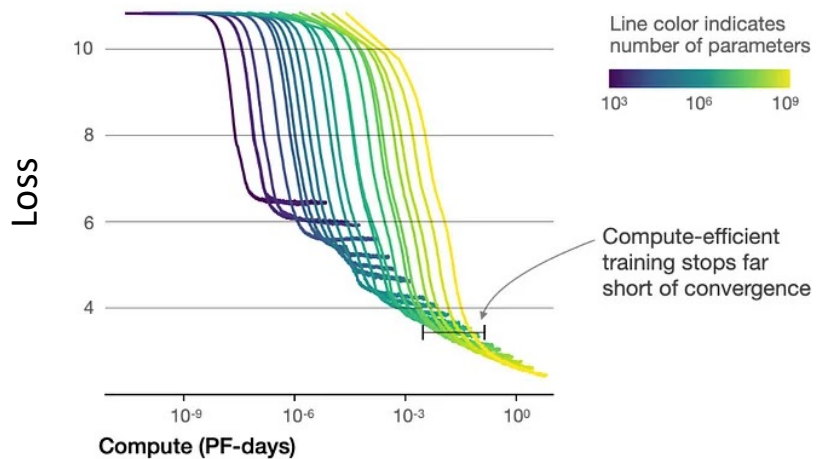
Take pre-trained models (done by few companies that can afford it) and adapt (fine-tune) that model to your tasks → This takes much less resources (few GPUs, few days, ~100 kWh)!

Scaling Laws for Foundation Models



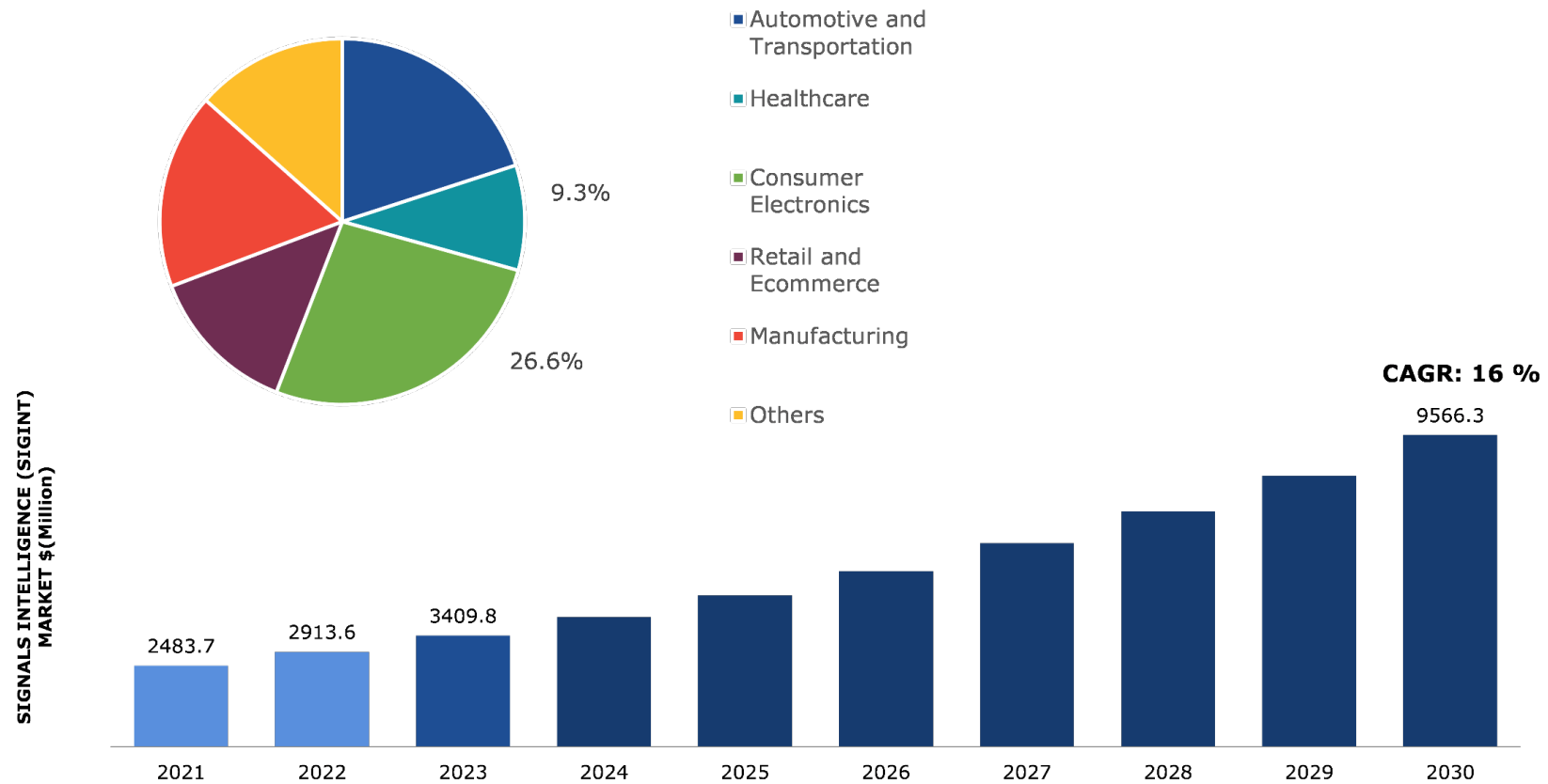
Language Model Scaling Laws and GPT-3

The optimal model size grows smoothly
with the loss target and compute budget



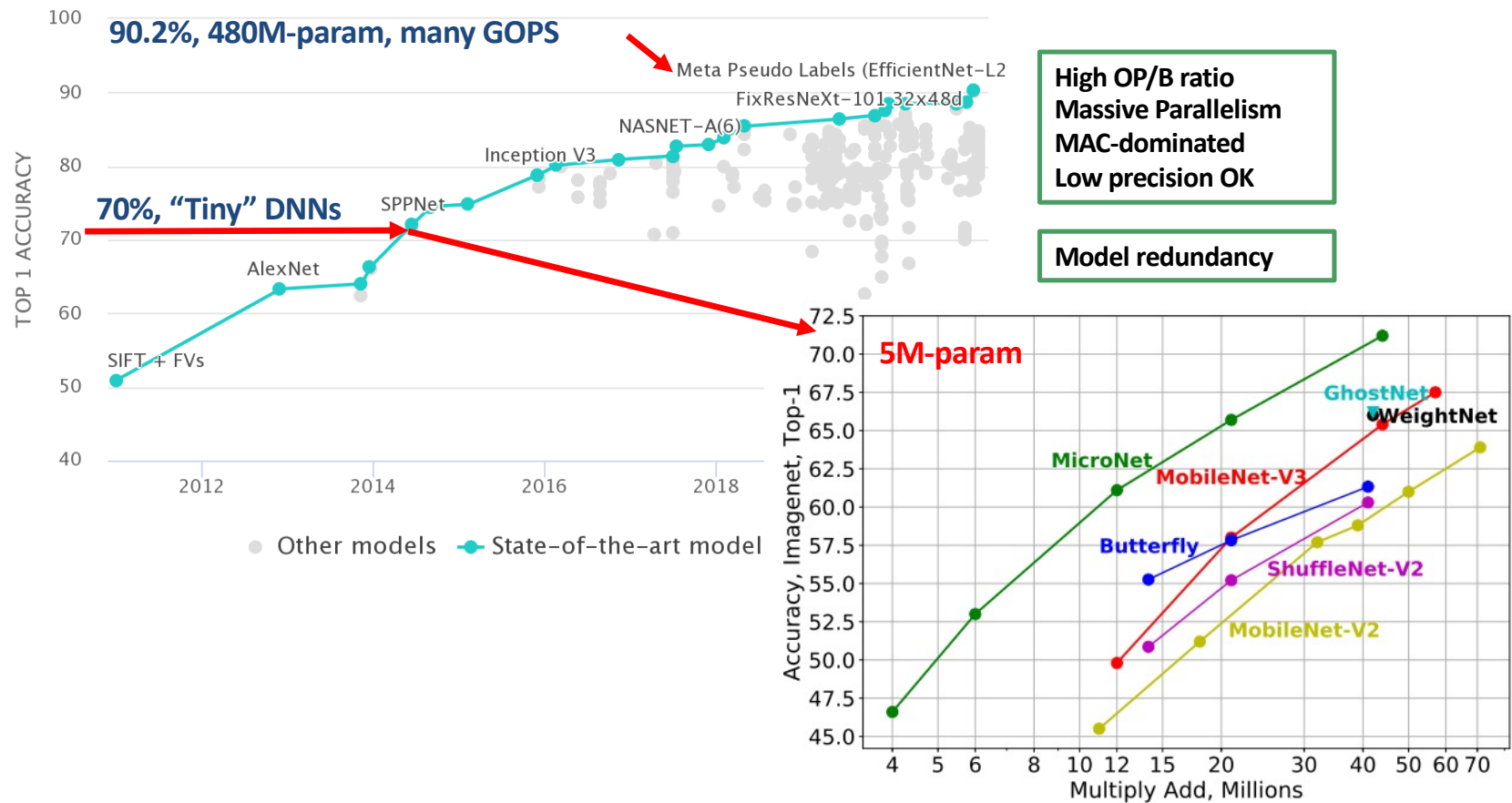
ML in the cloud → ML at the edge

[research dive 22]

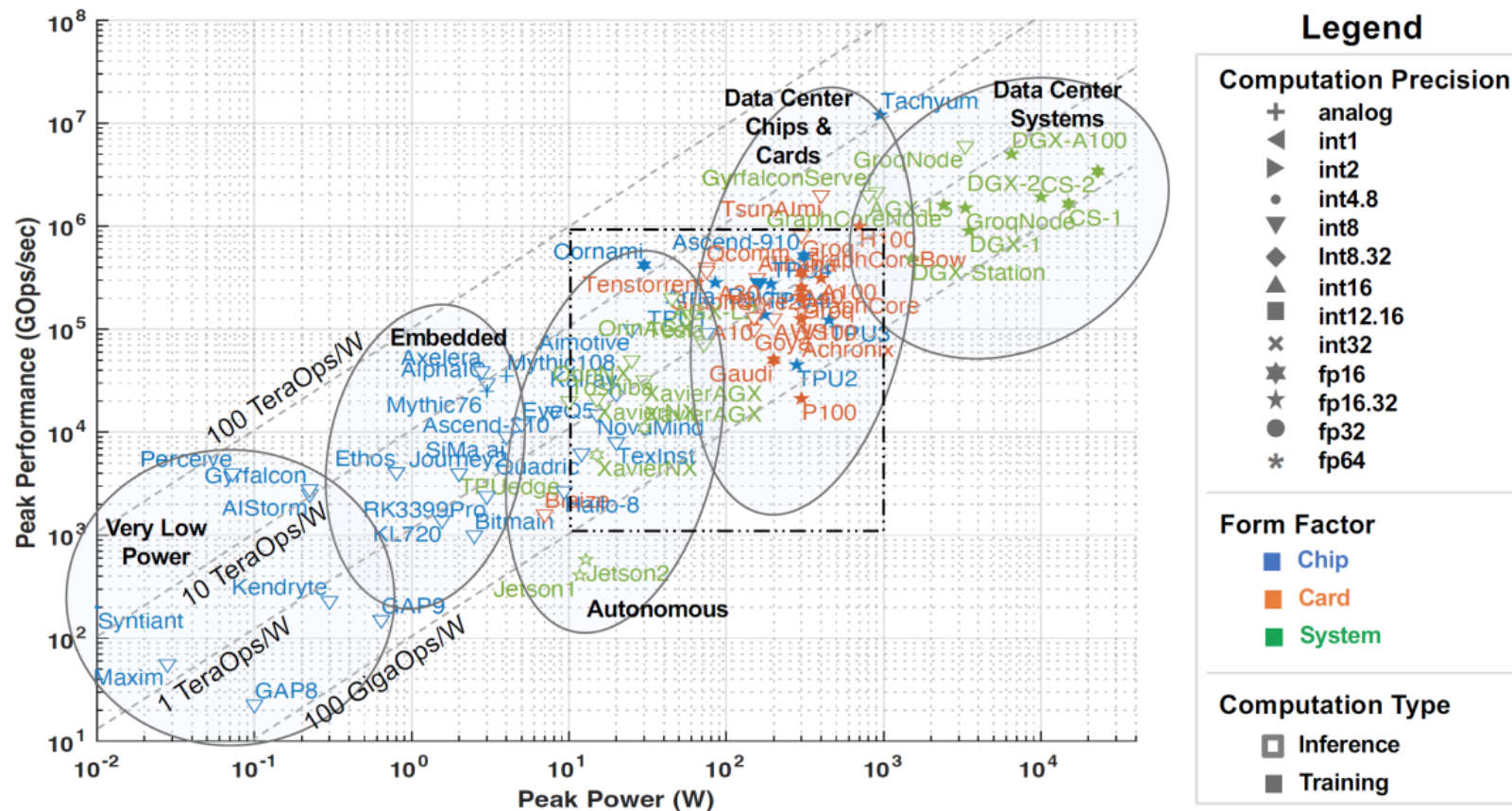


AI Workloads - DNNs

H Pham 2021(Google) arXiv:2003.10580v3



Machine Learning Hardware Frenzy



<https://arxiv.org/pdf/2210.04055.pdf>

Also refer to: <https://arxiv.org/abs/2306.15552> (2023)